ArrayList

Lecture 17

Based on Slides of Dr. Norazah Yusof

The `ArrayList` Class

- Similar to an array, an `ArrayList` allows object storage
- Unlike an array, an `ArrayList` object:
  - Automatically expands when a new item is added
  - Automatically shrinks when items are removed

- Requires:

```
import java.util.ArrayList;
```

Creating and Using an `ArrayList` and adding items using `add()` method

- Create an `ArrayList` object with no-args constructor
  - `ArrayList townList = new ArrayList();`
- To add element to the `ArrayList`, use the `add` method:
  - `townList.add("Kangar");`
  - `townList.add("Alor Setar");`
- To get the current size, call the `size` method
  - `townList.size();  // returns 2`
- Example: Lab 6 – Exercise 1 – Question 3

Accessing & Removing items in an `ArrayList`

- To access items in an `ArrayList`, use the `get` method as follows:

```
townList.get(1); // 1 is the index of the item to get.
```

- A loop is used in the following statement to access every element in the `ArrayList` named `townList`.

```
for(int i=0;i < townList.size();i++)
        System.out.print(townList.get(i)+" ");
```

- To remove items in an `ArrayList`, use the `remove` method

```
townList.remove(1); //This statement removes the second item.
townList.remove("Penang"); //This statement removes the item
                           // with the value "Penang".
```

Adding and replacing existing items using two argument method

- The `ArrayList` class's `add` method with one argument adds new items to the end of the `ArrayList`

- To insert items at a location of choice, use the `add` method with two arguments:

```
townList.add(6, "Shah Alam");
```

  This statement inserts the `String` "Shah Alam" at index 6

- To replace an existing item, use the `set` method:

```
townList.set(1, "Muar");
```

  This statement replaces the value at index 1 with “Muar”

Using `toString()` method

- The `ArrayList` class's `toString` method returns a string representing all items in the `ArrayList`

```
System.out.println(townList);
```

This statement yields :

```
[Muar, Alor Setar]
```

Using a Cast Operator with the `get` Method

- An `ArrayList` object is not typed
- To retrieve items from an `ArrayList`, you must cast the item to the appropriate type

```
ArrayList nameList = new ArrayList();
townList.add("Kluang"); // Inserts an item
String str = (String)townList.get(0);
```

- Try `get` without the cast to see the effect.

Using `ArrayList` as a Generic Data Type

- You can create a type-safe `ArrayList` object by using generics.
- For example an `ArrayList` object for `String`:

```
ArrayList<String> nameList = new
  ArrayList<String>();
```

- The `get` method no longer requires casts to work.

ArrayList Methods

- add(Object o) – add new object into ArrayList
- get(int index) - retrieves object reference from ArrayList *index* position
- size() - returns ArrayList size
- remove(int index) - removes the element at the specified position in this list. Shifts any subsequent elements to the left and returns the element that was removed from the list.
- indexOf(Object o) - finds the index in this list of the first occurrence of the specified element
- clear() - removes all of the elements

Exercises

- Do Exercise 1:
  - Question 1, page 115
  - Question 2, page 116
  - Question 3, page 117

Question 1, page 115

```
import java.util.ArrayList;

public class CircleArrayList {
    private double radius;

    public CircleArrayList(double radius)  {
        this.radius=radius;
    }

    public void setRadius() {
         this.radius=radius;
    }

    public double getRadius() {
         return radius;
    }

    public double getArea()  {
         return (Math.PI*radius*radius);
    }
}
```

Question 1, page 115

```
import java.util.ArrayList;

public class TestCircleAL {
  public static void main(String[] arg)       {
    ArrayList cList = new ArrayList();

    cList.add(new CircleArrayList(2.3));
    cList.add(new CircleArrayList(3.3));

    System.out.print("The area : %2.2f " +
                     ((CircleArrayList)cList.get(0)).getArea());
  }
}
```

Question 2, page 116

```
import java.util.ArrayList;
ArrayList student = new ArrayList();

student.add("Siti Rahimah");
student.add("Robert Lau");

System.out.print(student);
Student.set(1, "Muhammad");
```